Evenson's Mathematical Workshop

Evenson Auguste

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Introduction

Evenson's Mathematical Workshop is an evolving collection of mathematical problems and solutions written in the spirit of mathematical competitions and classical problem solving.

The goal of this project is to gather and preserve the problems that I solve and publish over time on my personal blog:

https://evenson0.github.io

Rather than letting each problem remain isolated in a blog post, this book collects them into a single workshop of ideas where arguments, techniques, and observations can be refined and organized.

The problems presented here cover various areas of elementary mathematics, especially combinatorics, algebra, and number theory. Each problem is accompanied by a concise solution emphasizing the key mathematical idea.

The collection is intended to grow gradually over time as new problems are solved and published.

Problem 1. *In a round-robin tournament with n players $P_1, P_2, \ldots, P_n$, each player plays exactly one match against every other player.*

There are no ties, so every match results in one win and one loss.

Let W_k denote the number of wins of player P_k, and let L_k denote the number of losses of player P_k.

Show that

$$\sum_{k=1}^{n} W_k^2 = \sum_{k=1}^{n} L_k^2.$$

(Combinatorics, Graph Theory)

Solution. *The identity can be written*

$$\sum_{k=1}^{n} (W_k^2 - L_k^2) = 0$$

or

$$\sum_{k=1}^{n} (W_k - L_k)(W_k + L_k) = 0.$$

For each player we have

$$W_k + L_k = n - 1,$$

since each player plays against the other $n-1$ players.

Thus

$$(n-1) \sum_{k=1}^{n} (W_k - L_k) = 0.$$

Since $n - 1 \neq 0$, we must have

$$\sum_{k=1}^{n} (W_k - L_k) = 0.$$

Equivalently

$$\sum_{k=1}^{n} W_k = \sum_{k=1}^{n} L_k.$$

Every match produces exactly one win and one loss, so the total number of wins equals the total number of losses.

Therefore

$$\boxed{\sum_{k=1}^{n} W_k^2 = \sum_{k=1}^{n} L_k^2}.$$

Problem 2. *Let $P(x)$ be a polynomial of degree n satisfying*

$$P(k) = k \quad \text{for } k = 1, 2, \ldots, n$$

and

$$P(0) = 1.$$

Find

$$P(-1).$$

(Algebra, Polynomials)

Solution. *Define*

$$Q(x) = P(x) - x.$$

Then $Q(x)$ has roots

$$1, 2, \ldots, n.$$

Hence

$$Q(x) = c \prod_{k=1}^{n} (x - k)$$

for some constant c.
Thus

$$P(x) = x + c \prod_{k=1}^{n} (x - k).$$

Using $P(0) = 1$ gives

$$1 = c(-1)^n n!$$

so

$$c = \frac{(-1)^n}{n!}.$$

Therefore

$$P(-1) = -1 + \frac{(-1)^n}{n!} \prod_{k=1}^{n} (-1 - k).$$

But

$$\prod_{k=1}^{n} (-1 - k) = (-1)^n (n + 1)!.$$

Thus

$$P(-1) = -1 + (n + 1) = n.$$

Hence

$$\boxed{P(-1) = n}.$$

Problem 3. *Find the smallest positive integer n such that every digit of $15n$ is either 8 or 0.*

(Number Theory, Divisibility)

Solution. *Let*

$$N = 15n.$$

Since N is divisible by 15, it must be divisible by both 3 and 5.

Because N is divisible by 5, its last digit must be 0 or 5. The digits are restricted to 0 and 8, so the last digit must be 0.

If N contains k digits equal to 8, then the sum of its digits is

$$8k.$$

For divisibility by 3 we require

$$8k \equiv 0 \pmod 3.$$

Since $8 \equiv 2 \pmod 3$, this implies

$$2k \equiv 0 \pmod 3.$$

Thus k must be a multiple of 3.

The smallest possibility is $k = 3$, giving

$$N = 8880.$$

Hence

$$n = \frac{8880}{15} = 592.$$

Therefore

$$\boxed{592}.$$

Remark. *If k denotes the number of digits equal to 8 in N, the condition $8k \equiv 0 \pmod 3$ implies that k must be a multiple of 3.*